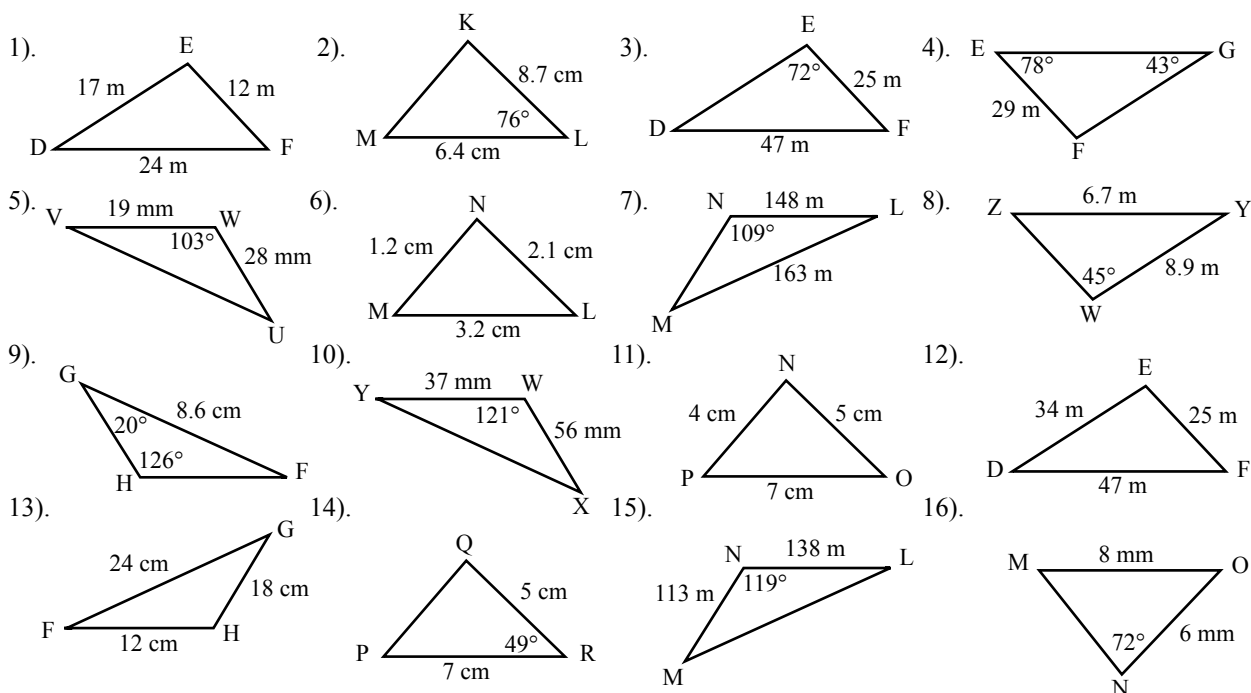




The Sine and Cosine Rules 2



Using the most appropriate rule, find **all** the missing lengths and angles in each diagram. Give the answer to 1 decimal place. Diagrams not to scale.



Worded Questions

- 1). Town B is 45 km due North of Town A. Town C is 38 km from Town A in a direction of 062° . Calculate the distance of Town B from Town C.
- 2). Two ships leave port at the same time. Ship A sails on a bearing of 054° for 50 km. Ship B sails on a bearing of 110° for 75 km. What is the distance between Ship A and Ship B?
- 3). Two ships leave port at the same time. Ship A sails on a bearing of 037° for 120 km. Ship B sails on a bearing of 160° for 85 km. What is the distance between Ship A and Ship B?
- 4). Two planes leave the airport at the same time. The first plane travels on a bearing of 040° at a speed of 260 km/h. The second plane travels on a bearing of 142° at a speed of 450 km/h. How far are they apart after 2 hours?
- 5). Two aircraft leave the airport at the same time. Aircraft A flies on a bearing of 012° for x km. Aircraft B flies on a bearing of 077° for 123 km. The distance between the two aircraft now is 160 km. What is the distance, x , that aircraft A has flown?
- 6). Two tanks start from base in a desert. Tank A drives on a bearing of 230° for 12 km and Tank B drives on a bearing of 268° for y km. The Tanks are now 33 km apart. Find the distance, y , that Tank B has driven.
- 7). Two ships leave port at the same time. One sails on a course of 099° at a speed of 18 km/h and the other sails on a course of 243° at a speed of 12 km/h. Find the distance between them after 30 minutes.
- 8). A man prospecting for oil in the desert drives 46 km on a bearing of 034° and then 38 km on a bearing of 164° .
 - a). How far from his base is he now?
 - b). On what bearing must he drive to return to his base?

